

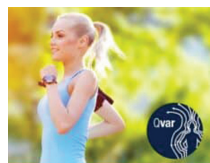
and PTCs. Therefore, it can enhance the safety of the system while reducing the cost and size of the device. The e-Fuse can reduce the design-in phase, PCB space requirements, BoM cost, and time-to-market compared to typical discrete solutions. These semiconductor based electronic fuses offer various safety features, such as overvoltage, overcurrent, short-circuit and inrush current protection. Moreover, these feature reverse current blocking, undervoltage lockout, soft start, and overtemperature protection with real-time diagnostics. The highly integrated silicon based fuse comes with real-time diagnostics and provides adjustability, flexibility, and reliable protection features in highly compact form factors that save PC board space. The eFuse is suitable for use in various applications, including portable electronics, wearables, enterprise servers, and power tools.

Littelfuse

<https://www.littelfuse.com>

Smart accelerometers

STMicroelectronics has launched three new accelerometers—the LIS2DU12, LIS2DUX12, and LIS2DUXS12—with advanced processing



engines built-in to extend sensor autonomy, enabling systems to respond more quickly

to external events while lowering power consumption. LIS2DUX12 and LIS2DUXS12 leverage ST's third-generation MEMS technology, adding programmable capabilities, including machine-learning core, advanced finite state machine, and an enhanced pedometer. LIS2DU12 is an entry level accelerometer for less demanding applications. All three are equipped with the latest industry standard I3C interface. The three devices integrate the common digital features for detecting events, as well as an anti-aliasing filter for high accuracy at lower sam-

pling frequencies with performance benefits for accurate gesture detection at negligible power consumption. The smart accelerometers provide context sensing for wearable devices, True Wireless Stereo (TWS) speakers and earbuds, smartphones, hearing aids, game controllers, smart watches, asset trackers, robotic appliances, and IoT devices.

STM

<https://www.st.com>

Low-power wireless SoCs

nRF54H20 is the first product in Nordic's nRF54 series, which has fourth-generation low-power wireless SoCs for IoT applications. The nRF54 series introduces a new hardware architecture fabricated on the GlobalFoundries 22FDX leading process node.



nRF54H20 is capable of supporting Bluetooth 5.4 and future Bluetooth specifications, plus LE Audio, Bluetooth

mesh, Thread, Matter, etc. It boasts multiple Arm Cortex-M33 processors running at speeds up to 320MHz and multiple RISC-V coprocessors. Each processor is optimised for a specific type of workload. nRF54H20 features multiple digital and analogue interfaces, including a high-performance external memory interface (400MBps), high-speed USB (480Mbps), two I3C peripherals, a CAN FD controller, and a 14-bit ADC. The low powered SoC can enhance the runtime of battery powered applications and features cryptographic accelerators that are hardened against side-channel attacks and tamper sensors that detect an attack in progress and take appropriate action. The SoC is suitable for a wide range of applications, including use in wearables and hearables, and is also suitable for applications demanding complex machine learning (ML) and support for sensor fusion at the edge.

Nordic Semiconductor

<https://www.nordicsemi.com>

LED driver ICs

BD83A04EFV-M, BD83A14EFV-M, and BD82A26MUF-M are 4ch/6ch LED driver ICs designed for medium- to large-size automotive displays in car infotainment and instrument clusters. The ICs developed by ROHM can eliminate flickering of LCD display while changing the display brightness from low to high. The ICs combine proprietary low power technology,



which reduces the power consumption by 20%. The flickering of LCD display's general

PWM dimming is reduced, improving system reliability. The company has also released four reference designs along with PCB design data, SPICE models, and EMC reports to simplify the job of design engineers, while also lowering the cost of speeding up the time to market. The LED drivers are suitable for use in various applications, including instrument clusters, car infotainment systems, heads-up displays, and electronic mirrors.

ROHM

<https://www.rohm.com>

EMBEDDED

Long-range transceiver modules

The RYLR684 and RYLR689 from Reyax Technology are cost-effective LoRa and (G)FSK (Gaussian frequency-shift keying) transceiver modules



that are suitable for long-range wireless applications. They are

based on a Semtech LLCC68 chip and have a programmable transmission power level from -20dBm to 22dBm. This allows the designers to optimise the range and power of the transmitter for each use case, thus reducing the energy consumption of battery-powered devices. The modules have been launched to replace the traditional